

MediConnect

AI-Based Symptom Checker & Doctor Recommendation System

with Integrated Medicine Reminder

Software Requirements Specification (SRS)

Document Version 1.0

Prepared by

Student ID	Name
24101307	Shibly Nomany Khan
24341202	Nurunnahar Nur
23201079	Nafisa Tahsin
23101373	Taufiqur Rahman

July 2026

Table of Contents

1. Introduction.....	3
1.1 Project Type.....	3
1.2 Purpose.....	3
1.3 Target Users.....	3
1.4 Definitions, Acronyms & Abbreviations.....	3
2. Feature Overview.....	3
3. Functional Requirements.....	4
3.1.1 User Management & Authentication.....	4
3.1.2 AI-Assisted Symptom Checker.....	4
3.1.3 Doctor & Hospital Discovery.....	5
3.1.4 Health Reports & Records.....	5
3.1.5 Medicine Reminder.....	5
3.1.6 Communication & Engagement.....	5
3.1.7 Administration.....	6
3.1.8 Sample Rule-Base Logic (Illustrative).....	6
4. Non-Functional Requirements.....	6
5. Class Diagram.....	7
6. Tools and Technologies.....	8
6.1 Frontend.....	8
6.2 Backend.....	8
6.3 Database.....	8
6.4 Other Tools & Services.....	8
7. Compatibility with System Environment / OS.....	8
8. Implementation.....	8
9. Challenges.....	9
10. Conclusion.....	9

1. Introduction

1.1 Project Type

MediConnect is a web-based healthcare support application that combines an AI-style, rule-based symptom checker with doctor and hospital recommendation, appointment booking, health-report generation, and a medicine reminder module. It is designed as a lightweight, easy-to-implement decision-support tool rather than a diagnostic medical device — its rule-based engine maps common symptom combinations to likely conditions and the appropriate specialist, without requiring a trained machine-learning model.

1.2 Purpose

The purpose of this document is to define the functional and non-functional requirements, architecture, and scope of MediConnect so that the development team, evaluators, and future contributors share a single, unambiguous reference. The system aims to help users quickly understand what their symptoms might indicate, which type of specialist to consult, where to find nearby care, and to support ongoing treatment through medicine reminders and health-report tracking.

1.3 Target Users

- Patients / General Users: people experiencing symptoms who want quick guidance on possible causes, the right specialist to see, nearby facilities, and help remembering medication.
- Doctors: healthcare providers who receive shared health reports, add diagnoses/prescriptions, and manage their availability.
- Administrators: manage the doctor/hospital directory, maintain the symptom-disease-specialist rule-base, and monitor system-wide analytics.

1.4 Definitions, Acronyms & Abbreviations

Term	Meaning
SRS	Software Requirements Specification
FR / NFR	Functional Requirement / Non-Functional Requirement
OTP	One-Time Password
RBAC	Role-Based Access Control
EMR	Electronic Medical Record
Rule Engine	A simple if/then mapping (e.g., Fever + Cough → Flu) used instead of a trained ML model

2. Feature Overview

MediConnect implements 26 core features spanning symptom checking, care discovery, reporting, reminders, and administration:

1. AI-assisted (rule-based) symptom checker	16. Mood & emotion tracker with trend view
2. Disease prediction from symptom combinations	17. Mental-wellbeing check-ins (Depression / Anxiety / OCD screening aid)
3. Expanded disease coverage incl. Depression, Anxiety, OCD, PCOS, Diabetes indicators	18. Menstrual cycle & flow tracker
4. Specialist / doctor-type recommendation	19. Cycle & fertile-window prediction, PCOS pattern flagging
5. Severity / urgency indicator	20. Diet & nutrition recommendation and meal logging
6. Nearby hospital & clinic finder (location-based)	21. Exercise recommendation, workout logging & rep/calorie count
7. Doctor & hospital filtering by specialty and distance	22. Blood pressure log with trend charts & abnormal-reading alerts
8. Doctor profile & availability display	23. Unified wellness dashboard (mood, cycle, diet, exercise, BP, symptoms)
9. Appointment booking, rescheduling & cancellation	24. Role-based dashboards (Patient / Doctor / Admin)
10. Automated health report generation (PDF)	25. Profile & medical history management
11. Health report history & symptom log	26. Doctor rating & feedback system
12. Report sharing with doctors	27. Educational health tips based on diagnosis or wellness data
13. Doctor diagnosis notes & prescriptions	28. Admin panel for hospitals, doctors & rule-base management
14. Medicine reminder creation (dosage, time, frequency) and recurring & auto-expiring medicine schedules	29. Emergency contact quick-access
15. Automated reminder notifications (push/SMS/email) and mark reminder as taken / skipped / snoozed	

3. Functional Requirements

3.1.1 User Management & Authentication

ID	Requirement Description
FR-1	The system shall allow users to register and log in using email and password, and shall route each user to a role-based dashboard (Patient, Doctor, or Admin) with access limited to their role.
FR-2	The system shall allow users to update their profile, medical history, and contact details, and to securely reset their password via an email link.

3.1.2 AI-Assisted Symptom Checker & Disease Identification

ID	Requirement Description
FR-3	The system shall allow patients to enter symptoms via a checklist or free-text search.
FR-4	The system shall apply a rule-based engine (symptom-to-disease mapping) to suggest possible conditions (e.g., Fever + Cough → Flu; Chest Pain → Cardiac concern).
FR-5	The system shall recommend the appropriate medical specialist based on the predicted condition (e.g., Skin Rash → Dermatologist).
FR-6	The system shall display a confidence/severity indicator (Low / Moderate / Urgent) alongside each suggestion.
FR-7	The system shall maintain an editable, extensible rule-base (symptom-disease-specialist mapping) manageable by the Admin, covering common physical conditions as well as mental-health and reproductive-health patterns (e.g., Depression, OCD, PCOS) with appropriate disclaimers and a recommendation to seek professional care.
FR-8	The system shall log each symptom-check session to the patient's history for future reference.

3.1.3 Doctor & Hospital Discovery

ID	Requirement Description
FR-9	The system shall show a list of nearby hospitals/clinics based on the patient's location (GPS or manual entry).
FR-10	The system shall allow filtering of hospitals/doctors by specialty, distance, and rating.
FR-11	The system shall display doctor profiles including specialty, experience, and available time slots.
FR-12	The system shall allow patients to book, reschedule, or cancel appointments with a recommended specialist.

3.1.4 Health Reports & Records

ID	Requirement Description
FR-13	The system shall automatically generate a downloadable health report (PDF) after each symptom check, listing symptoms, possible conditions, and recommendations.
FR-14	The system shall allow patients to view their historical health reports and symptom logs.
FR-15	The system shall allow patients to share a generated health report directly with a chosen doctor.
FR-16	The system shall allow doctors to add diagnosis notes and prescriptions linked to a patient's report.

3.1.5 Medicine Reminder

ID	Requirement Description
FR-17	The system shall allow patients to create medicine reminders with name, dosage, time, and frequency.
FR-18	The system shall send automated notifications (push/SMS/email) at scheduled medicine times.
FR-19	The system shall allow patients to mark a reminder as taken, skipped, or snoozed.
FR-20	The system shall allow recurring reminders (daily/weekly/custom) and automatically stop them after a course ends.

3.1.6 Health & Wellness Tracking (inspired by Apple Health / Flo)

ID	Requirement Description
FR-21	The system shall allow patients to log daily mood and emotional state through short check-ins, and shall display mood trends over time.
FR-22	The system shall use mental-wellbeing check-in patterns (e.g., persistent low mood, anxiety, intrusive/repetitive thought patterns) to flag possible signs of conditions such as Depression, Anxiety, or OCD, and recommend consulting a mental-health professional — the feature is a screening aid, not a diagnosis.
FR-23	The system shall provide a menstrual cycle and flow tracker allowing patients to log period start/end dates, flow intensity, and related symptoms.
FR-24	The system shall predict the next expected menstrual cycle and fertile window based on logged cycle history, and flag irregular-cycle patterns that may indicate a condition such as PCOS.
FR-25	The system shall provide personalized diet/nutrition recommendations based on the patient's profile, goals, and any logged conditions, and allow logging of meals.
FR-26	The system shall recommend physical exercises appropriate to the patient's profile and goals, and allow logging of workouts including type, duration, repetition/rep count, and estimated calories burned.

FR-27	The system shall allow patients to log blood pressure readings (systolic/diastolic) and other basic vitals, display trend charts, and flag abnormal readings.
FR-28	The system shall present mood, cycle, diet, exercise, blood-pressure, and symptom-check data together in a unified wellness dashboard.

3.1.7 Communication & Engagement

ID	Requirement Description
FR-29	The system shall send appointment and follow-up notifications via SMS/email/push.
FR-30	The system shall allow patients to rate doctors and leave feedback after a consultation.
FR-31	The system shall provide short educational health tips relevant to the symptoms/disease identified or the wellness data logged.

3.1.8 Administration

ID	Requirement Description
FR-32	The system shall allow Admins to manage the list of hospitals, doctors, and specialties.
FR-33	The system shall allow Admins to manage/extend the symptom-disease-specialist rule-base, including mental-health and reproductive-health patterns.
FR-34	The system shall provide Admins with an analytics dashboard showing the most common symptoms, disease trends, and aggregated (anonymized) wellness trends.

3.1.9 Sample Rule-Base Logic (Illustrative)

The symptom checker does not use machine learning. It uses a simple, extensible lookup table of symptom combinations mapped to a likely condition and recommended specialist:

Symptom Combination	Possible Condition	Recommended Specialist
Fever + Cough	Flu / Common Cold	General Physician
Fever + Sore Throat + Body Ache	Possible Viral Infection	General Physician
Chest Pain	Possible Cardiac Issue	Cardiologist
Shortness of Breath + Leg Swelling	Possible Heart / Kidney Issue	Cardiologist / Nephrologist
Skin Rash	Dermatological Condition	Dermatologist
Persistent Headache + Blurred Vision	Possible Neurological Issue	Neurologist
Joint Pain + Swelling	Possible Arthritis	Orthopedic / Rheumatologist

Abdominal Pain + Nausea	Possible Gastric Issue	Gastroenterologist
Fatigue + Diarrhea	Possible Gastroenteritis	Gastroenterologist / General Physician
Frequent Urination + Excessive Thirst + Fatigue	Possible Diabetes Indicator	Endocrinologist
Persistent Sadness + Loss of Interest + Low Energy (≥ 2 weeks)	Possible Depression	Psychiatrist / Clinical Psychologist
Excessive Worry + Restlessness + Racing Heart	Possible Anxiety Disorder	Psychiatrist / Clinical Psychologist
Intrusive Thoughts + Repetitive/Compulsive Behaviors	Possible OCD	Psychiatrist / Clinical Psychologist
Irregular Periods + Weight Gain + Acne / Excess Hair Growth	Possible PCOS	Gynecologist / Endocrinologist
Severe Menstrual Pain + Heavy Flow	Possible Menstrual Disorder	Gynecologist

Note: This mapping is for demonstration only and is not a substitute for professional medical, psychiatric, or gynecological diagnosis. Mental-health and reproductive-health patterns (Depression, Anxiety, OCD, PCOS, etc.) are shown only as screening prompts to encourage the user to seek a qualified professional; the app will display this disclaimer, and mental-health check-ins will include a visible link to crisis/helpline resources.

4. Non-Functional Requirements

Category	ID	Requirement Description
Performance	NFR-1	The symptom checker shall return possible conditions and specialist recommendations within 2 seconds of submission.
Performance	NFR-2	Average page load time shall not exceed 3 seconds under normal network conditions.
Security	NFR-3	All client-server communication shall occur over HTTPS (SSL/TLS).
Security	NFR-4	Sensitive data (medical history, reports, credentials) shall be encrypted at rest.
Security	NFR-5	The system shall enforce role-based access control (RBAC) so users only access authorized data.
Reliability	NFR-6	The system shall maintain 99% uptime excluding scheduled maintenance.
Reliability	NFR-7	Medicine reminder notifications shall be delivered with at most a 1-minute delay from the scheduled time.
Usability	NFR-8	The interface shall be responsive and usable on desktop, tablet, and mobile screens.

Category	ID	Requirement Description
Usability	NFR-9	The symptom-entry flow shall require no more than 3 steps to reach a recommendation.
Maintainability	NFR-10	The rule-based engine shall be stored in a structured, editable format (e.g., JSON/DB) so new symptom-disease mappings can be added without code changes.
Scalability	NFR-11	The system architecture shall support horizontal scaling of the backend and database as the user base grows.
Compliance	NFR-12	The system shall follow basic healthcare data privacy practices (informed disclaimers, consent for data storage) consistent with HIPAA/GDPR principles.

5. Class Diagram

The diagram below shows the core classes of MediConnect and their relationships: User is specialized into Patient, Doctor, and Admin; Patient interacts with SymptomChecker, which maps symptoms to Disease/Specialist and suggests a Hospital; Patient also manages Appointment, MedicineReminder, HealthReport, and Feedback objects, with Notification supporting reminders and appointment alerts.

6. Tools and Technologies

6.1 Frontend

- React.js — component-based, responsive user interface
- HTML5 / CSS3 / Tailwind (or Bootstrap) — layout and styling
- Axios / Fetch API — communication with backend REST APIs

6.2 Backend

- Node.js with Express.js — REST API and business logic, including the rule-based symptom engine
- JSON-based rule configuration file / MongoDB collection for the symptom–disease–specialist mapping
- JWT — session/token-based authentication; OTP via Twilio/SendGrid or similar

6.3 Database

- MongoDB — storing users, appointments, health reports, reminders, and hospital/doctor directory

6.4 Other Tools & Services

- Google Maps / Places API — nearby hospital and clinic discovery
- PDF generation library (e.g., pdfkit / jsPDF) — health report export
- Push/SMS/email notification service — medicine reminders and alerts
- Git & GitHub — version control and collaboration
- Figma — UI/UX design and prototyping

7. Compatibility with System Environment / OS

MediConnect is a web application, so it does not depend on any specific operating system for end users. It runs inside standard modern web browsers (Chrome, Firefox, Edge, Safari) on Windows, macOS, and Linux desktops, and on Android/iOS mobile browsers thanks to the responsive React front end. The Node.js/Express backend is OS-agnostic and can be deployed on Windows, Linux, or containerized cloud platforms (e.g., AWS, Render, Vercel/Netlify for the frontend). MongoDB Atlas (cloud-managed) is used so the database layer is likewise independent of the host OS. The only environment requirement for end users is a browser with JavaScript enabled and an internet connection; location-based features additionally require GPS/location permission to be granted by the user.

8. Implementation

This section is intended to show one or two screenshots of the working application per feature, each with a one- to two-sentence description, captured from the team's Sprint 1 demo build. The placeholders below

indicate where each screenshot should be inserted; please replace them with actual captures from the running application before submission.

9. Challenges

- Designing a rule-base that covers enough symptom combinations to be useful without becoming unwieldy or misleading to maintain by hand.
- Making clear to users that suggestions are informational only and not a medical diagnosis, to avoid misuse or false reassurance.
- Reliable, timely delivery of medicine reminder notifications across push/SMS/email channels.
- Accurate, real-time nearby-hospital results depend on third-party location/maps APIs and data quality.
- Ensuring patient data (symptoms, reports, history) is stored and transmitted securely within a short development timeline.
- Coordinating a 4-person team across a compressed sprint schedule while keeping the rule-base, UI, and backend in sync.
- Keeping the UI simple and non-technical for patients while still surfacing enough detail (severity, specialist, next steps) to be useful.

10. Conclusion

MediConnect brings together a simple, rule-based symptom checker, specialist and hospital recommendation, appointment booking, automated health reporting, and medicine reminders into a single, easy-to-use healthcare web application. By avoiding the complexity of a trained machine-learning model in favor of a transparent, extensible symptom-to-specialist rule-base, the team can deliver a working, demonstrable product within the sprint timeline while still addressing a genuine, everyday healthcare need. The requirements captured in this document will guide the remaining sprints and serve as the basis for testing and evaluation.